

Diagram illustrating the contraction of two vertices, $A^{\sigma_i^\dagger}$ and A^{σ_i} , resulting in the identity operator I .

The left side shows an incoming line entering a vertex labeled $A^{\sigma_i^\dagger}$ (represented by a blue triangle pointing left). Two internal lines connect this vertex to a second vertex labeled A^{σ_i} (represented by a blue triangle pointing right). The top internal line has an arrow pointing right, and the bottom internal line has an arrow pointing left. An outgoing line exits the second vertex.

This is equal to the right side, which shows a single horizontal line with an arrow pointing right, labeled I .

$$A^{\sigma_i^\dagger} \text{---} A^{\sigma_i} = I$$